A Jump-Line Obstruction to Fixed-Point Preservation of Variability

Automatic Math Discovery Run `fa_banach_001`
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Status. Candidate substantial partial result, likely valid. This is an obstruction to a natural fixed-point invariance principle, not a full resolution of the source paper's open-ended request for some suitable assumptions.

Abstract

Hinz, Tölle, and Viitasaari ask for assumptions under which the integral process associated with a BV coefficient is itself variable, so that a standard fixed-point construction might replace their Doss transform. We show that the most immediate invariance statement fails in dimension two under strong baseline hypotheses. A bounded, symmetric, uniformly positive definite, piecewise-constant coefficient can satisfy their determinant, curl-free-inverse, and angle conditions, while a smooth driver with both coordinates active sends an (s, p)-variable input path exactly onto the coefficient's jump line. The output is then not variable for any positive variability exponent. The example works for every prescribed finite p. It shows that a successful fixed-point theorem needs an output-side transversality or occupation-energy hypothesis, not only coefficient nondegeneracy and input variability. A short scalar positive result is added to illustrate the missing mechanism.

1 Source problem and scope

The source is M. Hinz, J. M. Tölle, and L. Viitasaari, *Variability of paths and differential equations with BV-coefficients* [1]. In Section 1.2, PDF page 6, after describing their Doss-transform existence theory, the authors state that the main open problem is to prove, under reasonable assumptions, that "the integral process itself will be variable." Their goal is to make standard fixed-point arguments available.

Because "reasonable assumptions" are deliberately unspecified, no example can refute the question as written. The result below instead tests a strong and natural baseline: bounded BV coefficients satisfying all three structural assumptions used by the source's multidimensional Doss theory, smooth drivers, and variable input paths. It proves that these conditions do not make the raw Picard map invariant on variability classes.

2 Definitions

Let $\varphi \in BV_{\mathrm{loc}}(\mathbb{R}^n)$, $s \in (0, 1)$, and $p \in [1, \infty]$. Following [1, Definition 2.1], a continuous path $X : [0, T] \to \mathbb{R}^n$ is (s, p)-variable with respect to φ if there is a relatively compact open neighborhood U of $X([0, T])$ such that

$$t \longmapsto \int_U |X_t - z|^{-n+1-s} \, \|D\varphi\|(dz) \quad \text{belongs to } L^p(0, T).$$

fractional Brownian motions Y in $\mathbb{R}^n$ with $n \geq 2$ and Hurst index $H > \frac{1}{2}$. One can regard Corollary 3.26 as (a partial) extension of the probabilistic [89, Theorem 2.1]. Our fourth main result is a related uniqueness result, Theorem 3.28. It shows that Assumptions 3.12 and 3.15 guarantee uniqueness in the class of variability solutions.

It would be desirable to replace the Doss transform by standard fixed point arguments. The main open problem to be settled is to prove that — under reasonable assumptions — the integral process itself will be variable. Another goal for future research is to target equations that, in addition to a BV_{loc}-diffusion coefficient, involve a drift vector field of low regularity. First results on variability and compositions involving discontinuous paths can be found in [55].

1.3 Structure of the article

The structure of the article is as follows: In Section 2 we introduce the notion of (s,p)-variability, define compositions $\varphi(X)$ ($\sigma(X)$, respectively), and state our results on existence and properties of (1.1). We also provide a change of variable formula and a result

Figure 1: The source statement on PDF page 6. The image is a direct crop of the source paper, not a retyped substitute.

For a matrix coefficient $\sigma = (\sigma_{jk})$, variability is required componentwise; we write $X \in \mathsf{V}(\sigma, s, p)$. The source abbreviation $\mathsf{V}(\sigma, s)$ means $p = 1$.

For a fixed starting point x_0 and driver $Y = (Y^1, Y^2)$, the Picard-type integral map is

$$(\mathcal{T}X)_t = x_0 + \int_0^t \sigma(X_u)\, dY_u.$$

In our example both paths are regular enough that this is simply the ordinary Riemann–Stieltjes integral.

3 Idea of the proof

Variability measures how the occupation of a path interacts with the gradient measure of the coefficient. For a coefficient jumping across a line, a path that leaves the line like $t \mapsto t^\alpha$ sees a potential of order $t^{-\alpha s}$, which is integrable after choosing $\alpha s p < 1$. We choose the two constant coefficient matrices on opposite sides of the line so that they both send the driver velocity $(1,1)$ to the tangent vector $(1,0)$. Consequently, although the input leaves the jump line transversally, the integral process runs along it. At a point of the line the relevant Riesz potential integrates $|r|^{-1-s}$ against line measure and is infinite.

The matrix choice is arranged through the inverses. Two positive-definite inverse matrices share their first column $(1,1)$ and differ only in the $(2,2)$ entry. Sharing that column forces both original matrices to send $(1,1)$ to $(1,0)$; varying only the inverse's $(2,2)$ entry also makes its rows distributionally curl-free across a horizontal jump.

4 The obstruction

Set $L = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = 0\}$ and define

$$B_+ = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B_- = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix},$$

$$A_+ = B_+^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \qquad A_- = B_-^{-1} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Let

$$\sigma(x_1,x_2)=\begin{cases}A_+, & x_2\ge 0,\\ A_-, & x_2<0.\end{cases}$$

The value on L is immaterial for the distributional derivative and for the integral below.

Theorem 1 (Jump-line failure of Picard invariance)**.** *The coefficient σ is bounded, belongs componentwise to $BV_{\mathrm{loc}}(\mathbb{R}^2)$, and is symmetric and uniformly positive definite. It satisfies the determinant, curl-free-inverse, and angle assumptions used in the source paper's multidimensional Doss construction.*

Fix $s\in(0,1)$ and $p\in[1,\infty)$. Choose

$$0<\alpha<\min\left\{1,\frac{1}{sp}\right\},\qquad X_t=(0,t^\alpha),\qquad Y_t=(t,t),\quad 0\le t\le 1.$$

Then $X\in C^\alpha([0,1];\mathbb{R}^2)\cap \mathsf{V}(\sigma,s,p)$. The driver is smooth and both its coordinates are nonconstant, but with $x_0=0$,

$$(\mathcal{T}X)_t=(t,0).$$

Moreover,

$$\mathcal{T}X\notin \mathsf{V}(\sigma,\rho,q)\quad \textit{for every } \rho\in(0,1) \textit{ and every } q\in[1,\infty].$$

Thus the raw integral map does not preserve even the union of positive variability classes. In the basic $p=1$ setting one may take $X_t=(0,t)$, a Lipschitz input, for every $s\in(0,1)$.

Proof. First, $\det A_+=1$ and $\det A_-=\frac{1}{2}$. The eigenvalues of A_+ are $(3\pm\sqrt{5})/2$, and those of A_- are $1\pm 1/\sqrt{2}$. Hence

$$\left(1-\frac{1}{\sqrt{2}}\right)|\xi|^2\le\langle\xi,\sigma(x)\xi\rangle\le\frac{3+\sqrt{5}}{2}|\xi|^2$$

for every ξ and almost every x. This proves bounded uniform positive definiteness and the determinant condition. It also gives the source angle condition, since

$$\langle\xi,\sigma(x)\xi\rangle\ge\frac{1-1/\sqrt{2}}{(3+\sqrt{5})/2}|\sigma(x)\xi|\,|\xi|.$$

The coefficient is constant on the two half-planes. Since

$$A_+-A_-=\frac{1}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix},$$

each distributional derivative $D\sigma_{jk}$ is a nonzero signed constant multiple of the normal vector measure on L. In particular, $\|D\sigma_{jk}\|=c_{jk}\mathcal{H}^1\lceil_L$ with $c_{jk}=1/2$.

The inverse field equals B_+ above L and B_- below it. The first row is the constant vector $(1,1)$. The second row is $(1,b(x_2))$, where b is a one-dimensional step function. Therefore both distributional row curls vanish:

$$D_1(\sigma^{-1})_{k2}-D_2(\sigma^{-1})_{k1}=0,\qquad k=1,2.$$

This is precisely the two-dimensional curl-free-inverse condition.

We next check input variability. Choose any bounded open neighborhood U of $X([0,1])$. Enlarging its line section only increases the potential, so for $t>0$ and every entry,

$$\begin{aligned}\int_U|X_t-z|^{-1-s}\,\|D\sigma_{jk}\|(dz)&\le\frac{1}{2}\int_{\mathbb{R}}(r^2+t^{2\alpha})^{-(1+s)/2}\,dr\\ &=C_st^{-\alpha s},\end{aligned}$$

where $C_s = \frac{1}{2}\int_{\mathbb{R}}(1+u^2)^{-(1+s)/2}\,du < \infty$. The potential is infinite at $t=0$, but that single time is irrelevant to $L^p(0,1)$. Since $\alpha sp < 1$,

$$\int_0^1 (C_s t^{-\alpha s})^p\,dt < \infty.$$

Thus $X \in \mathsf{V}(\sigma, s, p)$ componentwise.

For almost every $t > 0$, X_t lies in the upper half-plane, so $\sigma(X_t) = A_+$. Since

$$A_+\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix},$$

the ordinary Stieltjes integral gives

$$(\mathcal{T}X)_t = \int_0^t A_+\begin{pmatrix}1\\1\end{pmatrix}du = (t,0) =: Z_t.$$

Finally, let U be any relatively compact open neighborhood of the output segment $Z([0,1])$. For every $t \in [0,1]$, openness supplies $\varepsilon_t > 0$ such that $\{(r,0) : |r-t| < \varepsilon_t\} \subset U$. Hence for every $\rho \in (0,1)$ and every entry,

$$\int_U |Z_t - z|^{-1-\rho}\,\|D\sigma_{jk}\|(dz) \geq \frac{1}{2}\int_{t-\varepsilon_t}^{t+\varepsilon_t} |t-r|^{-1-\rho}\,dr = +\infty.$$

The potential is infinite at every time, so it belongs to no L^q space. This proves the claim. □

Remark 1 (Compatibility with the source integral regime)**.** *The Lipschitz path Y belongs to C^γ for every $\gamma < 1$. Given the chosen α and s, take $\gamma \in (1-\alpha s, 1)$. Then $\alpha s + \gamma > 1$, exactly the source condition ensuring existence of the Stieltjes integral for an $(s,1)$-variable input. Here the integrand is in fact constant almost everywhere, so the integral is classical.*

5 A scalar positive subcase

The obstruction is tangential: the output is forced to run inside a jump set. The following elementary observation records a setting in which this cannot happen.

Proposition 1 (Positive scalar time driver)**.** *Let $n = 1$, $Y_t = t$, and $\sigma \in BV_{\mathrm{loc}}(\mathbb{R}) \cap L^\infty(\mathbb{R})$. Suppose an input path X is such that the composition is well defined and, for some constants $0 < c \leq M < \infty$,*

$$c \leq \sigma(X_t) \leq M \qquad \text{for almost every } t \in [0,T].$$

Then

$$Z_t = x_0 + \int_0^t \sigma(X_u)\,du$$

belongs to $\mathsf{V}(\sigma, \rho)$ for every $\rho \in (0,1)$.

Proof. The path Z is absolutely continuous and $c(t-u) \leq Z_t - Z_u \leq M(t-u)$ for $u < t$. Thus it is increasing and its inverse on its image is $1/c$-Lipschitz. Let U be a relatively compact open neighborhood of $Z([0,T])$ and put $\mu = \|D\sigma\| \restriction_U$, a finite measure. By Tonelli and the monotone change of variables,

$$\begin{aligned}\int_0^T \int_U |Z_t - z|^{-\rho}\,\mu(dz)\,dt &\leq \frac{1}{c}\int_U \int_{Z_0}^{Z_T} |y-z|^{-\rho}\,dy\,\mu(dz)\\ &< \infty,\end{aligned}$$

because $\rho < 1$, the two integration sets are bounded, and $\mu(U) < \infty$. This is precisely $(\rho,1)$-variability. □

Proposition 1 is not offered as a solution to the general open problem. It instead exposes the missing feature: a quantitative crossing condition turns the occupation measure of the output into a controlled one-dimensional measure. In higher dimensions, uniform positive definiteness alone does not prevent tangency to a codimension-one jump set, as Theorem 1 shows.

6 Verification, limitations, and novelty

The proof has no computational dependency. Its checkable core consists of:

1. the displayed matrix inversions and eigenvalue bounds;
2. the identity $A_+(1,1)^\top = (1,0)^\top$;
3. the scaling $\int_{\mathbb{R}}(r^2+a^2)^{-(1+s)/2}dr = C_s a^{-s}$;
4. divergence of $\int_{-\varepsilon}^{\varepsilon}|r|^{-1-\rho}dr$.

The result does not assert that the source problem has a negative answer. It leaves open positive fixed-point results under output-transversality, occupation-density, stochastic nonconcentration, or locally bounded gradient potential assumptions. The finite-p strengthening also does not give an (s,∞)-variable input, because the common starting point lies on the jump line.

A bounded search on 26 August 2026 covered the run's four lightweight indexes, the exact source phrase and arXiv id, the authors' follow-ups arXiv:2105.06249 [2] and arXiv:2407.06907 [3], and close terms involving fixed points, integral-process variability, Riesz potentials, and jump lines. No matching statement was found. Since the observation is elementary and the search was not a systematic citation review, the correct status is *candidate substantial partial result, likely valid*, not a definitive novelty claim.

7 Human-review recommendation

The packet is suitable for focused human review. The most important semantic question is whether the source authors intended the "integral process" as the raw Picard image of arbitrary admissible inputs, as here, or only after restricting the candidate set by an additional output-side occupation condition. Under the raw-map interpretation, the obstruction is exact.

References

[1] M. Hinz, J. M. Tölle, and L. Viitasaari, *Variability of paths and differential equations with BV-coefficients*, Ann. Inst. H. Poincaré Probab. Statist. 59 (2023), 2036–2082; arXiv:2003.11698.

[2] M. Hinz, J. M. Tölle, and L. Viitasaari, *Sobolev regularity of occupation measures and paths, variability and compositions*, Electron. J. Probab. 27 (2022), 1–29; arXiv:2105.06249.

[3] M. Hinz, J. M. Tölle, and L. Viitasaari, *Variability and the existence of rough integrals with irregular coefficients*, arXiv:2407.06907v2 (2025).